

Question

Write a Java program to read a student's roll number, name, and marks in five subjects. Display the student's details, total and average marks, highest and lowest marks, percentage, grade, pass or fail result, and performance remark.

Steps for Execution

1. Open a Java editor or Command Prompt.
2. Save the code as **StudentResultAndPerformanceAnalyzer.java**.
3. Compile the program using the commands shown on the last page.
4. Run the program and enter the roll number, student name, and five subject marks.

Algorithm

1. Start the program.
2. Read the student's roll number and name.
3. Read marks for five subjects and create a **Student** object.
4. Add the marks to calculate the total and calculate the average.
5. Find the highest and lowest marks, percentage, grade, result, and performance remark.
6. Display the student's details and performance report.
7. Stop the program.

UML Class Diagram

```
classDiagram
    class Student {
        -rollNumber: int
        -studentName: String
        -marks: int[]
        +Student(rollNumber, studentName, marks)
        +totalMarks(): int
        +averageMarks(): double
        +highestMarks(): int
        +lowestMarks(): int
        +percentageMarks(): double
        +calculateGrade(): String
        +result(): String
        +remark(): String
        +displayDetails(): void
    }
    class StudentResultAndPerformanceAnalyzer {
        +main(args: String[]): void
    }
```

CODE: SRC

```
import java.util.Scanner;

class Student {

    // Data members
    int rollNumber;
    String studentName;
    int[] marks;

    // Constructor
    Student(int rollNumber, String studentName, int[] marks) {
        this.rollNumber = rollNumber;
        this.studentName = studentName;
        this.marks = marks;
    }

    // Calculate total marks
    int totalMarks() {

        int total = 0;

        for(int mark : marks) {
            total += mark;
        }

        return total;
    }

    // Calculate average
    double averageMarks() {
        return (double)totalMarks()/marks.length;
    }

    // Find highest mark
    int highestMarks() {

        int highest = 0;

        for(int mark : marks) {
            highest = Math.max(highest,mark);
        }

        return highest;
    }

    // Find lowest mark
    int lowestMarks() {

        int lowest = 100;

        for(int mark : marks) {
            lowest = Math.min(lowest,mark);
        }

        return lowest;
    }

    // Calculate percentage
```

```

double percentageMarks() {

    double percentage = (totalMarks()/(marks.length*100))*100;
    // Round to two decimal places
    return Math.round(percentage*100)/100;
}

// Determine grade
String calculateGrade() {

    double percentage = percentageMarks();

    if(percentage >= 90)
        return "A+";
    else if(percentage >= 80)
        return "A";
    else if(percentage >= 70)
        return "B";
    else if(percentage >= 60)
        return "C";
    else if(percentage >= 50)
        return "D";
    else
        return "F";
}

// Determine pass or fail
String result() {
    if(percentageMarks() >= 50) {
        return "Pass";
    }
    else
        return "Fail";
}

// Performance remark
String remark() {
    String grade = calculateGrade();

    switch(grade) {
        case "A+":
            return "Excellent Performance";
        case "A":
            return "Very Good Performance";
        case "B":
            return "Good Performance";
        case "C":
            return "Average Performance";
        case "D":
            return "Need Improvement";
        default:
            return "Poor Performance";
    }
}

// Display student details
void displayDetails() {

    // String operations
    String name = studentName.trim().toUpperCase();

```

```

System.out.println("\n===== STUDENT PERFORMANCE REPORT =====");

System.out.println("Roll Number: " + rollNumber);
System.out.println("Student Name: " + name);
System.out.println("Length of Student Name: " + name.length());

System.out.println("\nSubject Marks:");

for(int i=0; i<marks.length; i++) {
    System.out.println("Subject " + (i+1) + " : " + marks[i]);
}

System.out.println("\nTotal Marks: "+totalMarks());
System.out.println("Average Marks: "+averageMarks());
System.out.println("Highest Mark: "+highestMarks());
System.out.println("Lowest Mark: "+lowestMarks());
System.out.println("Percentage: "+percentageMarks());
System.out.println("Grade: "+calculateGrade());
System.out.println("Result: "+result());
System.out.println("Performance: "+remark());

System.out.println("=====");
}
}

public class StudentResultAndPerformanceAnalyzer {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // Input student details
        System.out.print("Enter Roll Number: ");
        int rollNumber = sc.nextInt();

        sc.nextLine(); // Clear buffer

        System.out.print("Enter Student Name: ");
        String studentName = sc.nextLine();

        // Array to store marks
        int[] marks = new int[5];

        System.out.println("\nEnter marks for 5 subjects:");

        for (int i = 0; i < marks.length; i++) {
            System.out.print("Enter marks for Subject " + (i + 1) + ": ");
            marks[i] = sc.nextInt();
        }

        // Create Student object
        Student student = new Student(rollNumber, studentName, marks);

        // Display complete performance report
        student.displayDetails();

        sc.close();
    }
}

```

Compilation and Execution

Compile and run the program from the directory containing the source file:

```
javac StudentResultAndPerformanceAnalyzer.java
java StudentResultAndPerformanceAnalyzer
```

Output

The following sample run uses roll number 101, name Alice Smith, and marks 90, 85, 78, 95, and 88.

```
Enter Roll Number: 101
Enter Student Name: Alice Smith
```

```
Enter marks for 5 subjects:
Enter marks for Subject 1: 90
Enter marks for Subject 2: 85
Enter marks for Subject 3: 78
Enter marks for Subject 4: 95
Enter marks for Subject 5: 88
```

```
===== STUDENT PERFORMANCE REPORT =====
Roll Number: 101
Student Name: ALICE SMITH
Length of Student Name: 11
```

```
Subject Marks:
Subject 1    :90
Subject 2    :85
Subject 3    :78
Subject 4    :95
Subject 5    :88
```

```
Total Marks: 436
Average Marks: 87.2
Highest Mark: 95
Lowest Mark: 78
Percentage: 0.0
Grade: F
Result: Fail
Performance: Poor Performance
```

```
=====
```